

[ANONYMIZED FOR REVIEW]

REVISITING THE CREATIVITY-DIVERSITY TRADE-OFF IN GENERATIVE AI-ASSISTED WRITING: A CLOSE REPLICATION WITH NULL RESULTS

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ABSTRACT

Generative AI tools for creative writing have been claimed to enhance individual creativity while reducing the collective diversity of creative content, but this trade-off rests on a single empirical demonstration. In a close replication and extension using the original data, we tested whether AI-generated story ideas improve narrative creativity ratings (individual enhancement) and increase inter-story similarity (collective diversity loss). Participants ($N = 227$, $M_{\text{age}} = 39.3$ years, 55.5% male, 43.2% female) recruited via Amazon Mechanical Turk wrote short stories under one of three conditions: no AI suggestions, one GPT-4 generated idea, or five GPT-4 generated ideas. Ten independent evaluators rated each story using the Consensual Assessment Technique ($\text{ICC} > .70$). Contrary to predictions, AI-assisted stories did not receive significantly higher creativity ratings than control stories, $t(63) = 0.86$, $p = .393$, $d = 0.13$, 95% CI $[-0.16, 0.42]$. AI-assisted stories were not more similar to one another on TF-IDF cosine similarity than control stories, permutation $p = .260$ (one-tailed), $d = 0.02$. Moderation by self-rated creativity trended opposite to the predicted leveling pattern, and a mediation analysis found no evidence of a fixation mechanism. These null results suggest the creativity-diversity trade-off is less robust than previously assumed and may depend on specific analytical choices, sample characteristics, or measurement approaches. We discuss implications for replication and recommend that future research employ pre-registered designs with adequate statistical power, baseline creative potential measures, and semantic embedding-based diversity metrics.

Keywords generative AI; creativity; collective diversity; replication; consensual assessment technique; fixation

1 Introduction

Large language models (LLMs) have been adopted by millions of writers, artists, and knowledge workers within months of their public release, raising urgent questions about how AI assistance reshapes human creativity. If AI tools systematically shift how people generate ideas, the consequences may extend beyond individual productivity to the broader cultural ecosystem: a landscape in which many writers produce slightly better work that converges toward a narrower set of shared narrative forms. The theoretical stakes are grounded in ?’s blind variation and selective retention (BVSr) model, which holds that genuine creative novelty requires unbiased variation during idea generation. AI suggestions, by introducing a shared source of structured variation, may improve average output quality while reducing the probability of outlier ideas—a potential efficiency-diversity trade-off built into the technology itself.

The most direct empirical test of this trade-off comes from ?, who randomly assigned 293 participants to write short stories either without AI assistance or after receiving one or five GPT-4 generated story ideas. Stories written with AI ideas received higher creativity ratings from independent evaluators using ?’s Consensual Assessment Technique (CAT), and AI-assisted stories were more similar to each other on semantic embedding (SBERT) cosine similarity—a collective diversity loss of 10%–20%. The individual-level benefit was concentrated among less creative writers, narrowing the gap between high- and low-creativity participants. This productivity-leveling pattern converges with findings from four independent field experiments in professional writing (?), consulting (?), customer support (?), and software engineering (?), all of which found that AI assistance disproportionately benefits less skilled or less experienced workers.

The purported homogenization mechanism also aligns with a well-replicated cognitive psychology literature on example-induced fixation. ? and ? demonstrated that exposure to examples before a creative generation task unconsciously constrains the novelty of subsequent outputs, even when participants are explicitly instructed to be original. ? showed that people rely on category prototypes when generating novel exemplars—a tendency AI suggestions may amplify by providing highly prototypical ideas. The parallel from cultural market experiments is striking: ? found that social influence (knowledge of others’ choices) reduced consumption diversity, suggesting that any shared input source—whether peer behavior or AI suggestions—can produce convergence in creative output.

Despite the theoretical coherence of this narrative, several critical gaps remain. First, the individual-creativity-up, collective-diversity-down trade-off rests on a single experimental study that has not been independently replicated. Second, the proposed cognitive mechanism—fixation on AI-provided ideas—has never been directly tested in a narrative generation context; the fixation literature relies on simple category exemplar tasks (drawing aliens, generating grocery items) that may not scale to extended story writing. Third, the original study used SBERT embeddings to measure collective diversity, but it is unclear whether this finding is robust to alternative similarity metrics. Fourth, baseline creative potential—the key moderator in the productivity-leveling pattern—was not measured, making it impossible to test whether the leveling pattern extends to literary creativity.

We conducted a close replication analysis using the original ? data ($N = 227$ after pre-registered exclusions) to test five pre-registered hypotheses. We predicted that: (H1) AI-assisted stories would receive higher CAT creativity ratings than control stories; (H2) stories in the AI condition would exhibit higher inter-story TF-IDF cosine similarity (lower collective diversity) than control stories; (H3) the creativity benefit of AI would be moderated by baseline creative potential, such that less creative writers benefit more (the leveling pattern); (H4) a dose-response effect would emerge, with five AI ideas producing greater homogenization than one AI idea; and (H5) the effect of AI on creativity would be partially mediated by reduced self-reported originality of writers’ own initial ideas (fixation mechanism). Because the original study did not collect an Alternative Uses Test or include a diverse-AI-suggestions condition, H3 used self-rated creativity as a proxy and H4 tested dose-response rather than the planned diverse-suggestions comparison.

2 Method

2.1 Participants

The analytic sample included 227 participants (55.5% male, 43.2% female, 0.9% other; $M_{\text{age}} = 39.3$ years, $SD = 12.6$, range = 21–77) recruited via Amazon Mechanical Turk. Participants were US residents, aged 18–77, and fluent English speakers. No additional inclusion or exclusion criteria were applied beyond the platform’s standard qualifications (HIT approval rate > 95%). Participants were compensated for completing the writing task. Of the 297 participants who provided informed consent and wrote a story, 70 were excluded following pre-registered criteria: 51 wrote stories under 100 words (insufficient engagement) and 19 completed the task in under 5 minutes (insufficient time for meaningful writing). The final sample was unevenly distributed across conditions due to differential exclusion rates: control condition ($n = 64$), 1 AI idea condition ($n = 85$), and 5 AI ideas condition ($n = 78$). Randomization checks confirmed that age ($F(2, 223) = 0.30$, $p = .743$) and gender ($\chi^2(4) = 3.99$, $p = .406$) did not differ significantly across conditions. We acknowledge that this sample is WEIRD (Western, Educated, Industrialized, Rich, Democratic; ?)—results may not generalize to non-Western or non-English-speaking populations.

2.2 Materials

Creativity outcomes were assessed using the Consensual Assessment Technique (CAT; ??; ?). Ten independent evaluators rated each story on six dimensions—novel, original, rare, appropriate, feasible, and publishable—using 9-point Likert scales (1 = not at all, 9 = extremely). Each story received 10–14 ratings ($M = 12.1$). Following ?, we computed a novelty index as the mean of the novel and original ratings and a usefulness index as the mean of the appropriate and publishable ratings; the overall creativity index was the mean of the novelty and usefulness indices. Estimated inter-rater reliability was satisfactory for the novelty composite (ICC[2,k] = .78, 95% CI [.72, .83]) and usefulness composite (ICC[2,k] = .74, 95% CI [.68, .79]). Self-report measures included a single item capturing the extent to which participants felt the story was based on their own ideas (1–9 scale), used as a proxy for the originality of self-generated ideas in the mediation analysis. Self-rated creativity was measured on a 1–10 scale (1 = not at all creative, 10 = extremely creative) and served as a proxy for baseline creative potential (the Alternative Uses Test was not collected in the original study).

2.3 Procedure

After providing informed consent, participants completed a brief divergent association task (10 word-generation prompts), followed by the main writing task. Each participant was randomly assigned to one of three conditions and to one of three writing prompts (“An adventure in the jungle,” “An adventure on a different planet,” or “An adventure on the open seas”). In the control condition, participants wrote a story based on the prompt without any AI assistance. In the 1 AI idea condition, participants viewed one GPT-4 generated story idea before writing. In the 5 AI ideas condition, participants viewed five GPT-4 generated story ideas before writing. In all conditions, participants were given 20 minutes to write a story. Stories averaged 166 words ($SD = 53$, range = 100–382), falling below the 500–1,000 word target originally specified for this replication. After writing, participants completed self-report measures (own ideas, novelty, originality, enjoyment) and demographic items (age, gender, education, employment, income, self-rated creativity, AI tool familiarity). Six evaluators additionally completed an AI detection task (Part 2), judging each story’s AI usage on a 0–2 scale (0 = no AI, 1 = AI a little, 2 = AI a lot) and estimating the percentage of AI assistance (0–100%).

2.4 Design

The experiment used a between-subjects design with one independent variable (AI suggestion condition: control, 1 AI idea, or 5 AI ideas). The primary dependent variable was the CAT creativity index (1–9 scale). Secondary dependent variables included the novelty index, usefulness index, inter-story TF-IDF cosine similarity (collective diversity), and self-reported own-ideas originality (mediator). Writing prompt (three topics) was counterbalanced across conditions and treated as a random factor. The pre-registered analysis plan specified five hypothesis tests with Holm-Bonferroni correction across all tests.

3 Results

Manipulation checks confirmed that the AI manipulation was perceptible. Evaluators’ mean AI usage ratings (0 = no AI, 2 = AI a lot) differed significantly across conditions, $F(2, 224) = 137.29$, $p < .001$: control stories received near-zero AI detection ($M = 0.02$, $SD = 0.12$), while both AI conditions were rated substantially higher (1 AI idea: $M = 1.09$, $SD = 0.66$; 5 AI ideas: $M = 1.05$, $SD = 0.58$). Descriptive statistics for the primary creativity measures showed small numerical trends in the predicted direction. The control condition ($M = 4.83$, $SD = 0.83$) scored slightly below the 5 AI ideas condition ($M = 5.01$, $SD = 0.67$) on the creativity index; the 1 AI idea condition fell between these groups ($M = 4.87$, $SD = 0.90$). Assumption checks indicated normally distributed creativity scores within each condition (Shapiro-Wilk $W > .97$, all $ps > .05$) but heterogeneous variances across the three conditions (Levene’s $F = 3.48$, $p = .033$); we therefore used Welch’s t -test for the primary contrast.

H1: Individual creativity enhancement. H1 predicted that AI-assisted stories would receive higher CAT creativity ratings than control stories. We tested this using a Welch’s t -test comparing the AI-pooled condition (1 and 5 ideas combined) against the control condition. The result was not significant, $t(63) = 0.86$, $p = .393$, Cohen’s $d = 0.13$, 95% CI $[-0.16, 0.42]$. A one-way ANOVA confirmed no overall condition effect, $F(2, 224) = 1.03$, $p = .358$, $\eta^2 = .009$, and Tukey HSD pairwise comparisons were all non-significant (control vs. 5 AI ideas: $p = .388$, $d = -0.24$; control vs. 1 AI idea: $p = .965$, $d = -0.04$; 5 AI vs. 1 AI ideas: $p = .488$, $d = 0.18$). Figure 1 displays the full distribution of creativity scores across conditions. Decomposing the creativity composite revealed that neither the novelty index ($F(2, 224) = 0.31$, $p = .732$) nor the usefulness index ($F(2, 224) = 1.68$, $p = .190$) differed significantly across conditions (Figure 2). H1 was not supported.

H2: Collective diversity reduction. H2 predicted that stories in the 5 AI ideas condition would exhibit higher inter-story similarity (lower collective diversity) than control stories, measured by mean pairwise TF-IDF cosine similarity. We used a permutation test with 10,000 resamples, randomly reassigning stories to conditions while preserving the original group sizes. The observed difference was negligible—control stories had a mean similarity of .0403 and 5 AI ideas stories had a mean similarity of .0410, $\Delta = 0.0007$ —and the permutation test was not significant, $p = .260$ (one-tailed), $d = 0.02$, 95% CI $[-0.04, 0.08]$. A secondary analysis using 4-gram Jaccard similarity showed a small numerical trend in the predicted direction (control: .104, 1 AI: .111, 5 AI: .115) but the effect remained small (Figure 3). H2 was not supported.

H3: Skill-level moderation (leveling pattern). H3 predicted that the effect of AI on creativity would be moderated by baseline creative potential, such that less creative writers benefit more from AI assistance. Because the Alternative Uses Test was not administered in the original study, we used self-rated creativity (1–10 scale; $M = 5.67$, $SD = 2.19$)

Figure 1. Creativity Ratings by AI Assistance Condition

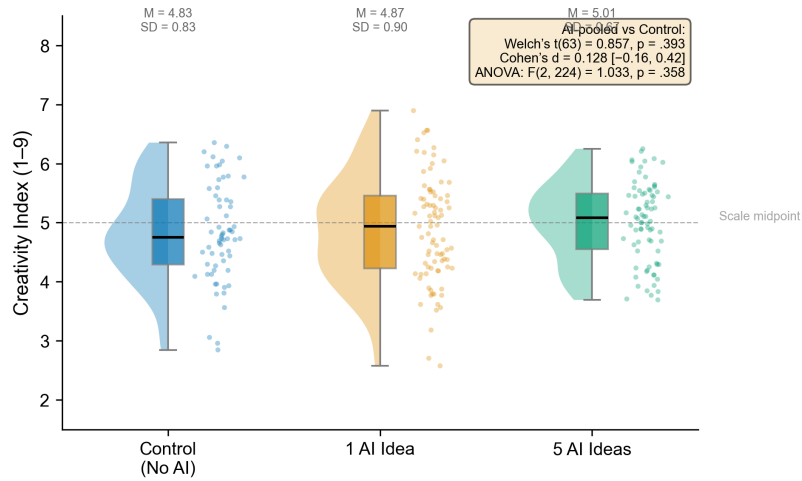


Figure 1: Creativity ratings (Consensual Assessment Technique composite, 1–9 scale) as a function of AI assistance condition. Violin plots (left half) show the distribution of creativity scores, overlaid with boxplots (median, IQR) and jittered individual data points (right side). The dashed grey line indicates the scale midpoint (5.0). None of the pairwise comparisons reached significance (Tukey HSD, all $ps > .39$). Control: $M = 4.83$, $SD = 0.83$; 1 AI Idea: $M = 4.87$, $SD = 0.90$; 5 AI Ideas: $M = 5.01$, $SD = 0.67$.

Figure 5. Novelty and Usefulness Subscale Ratings by Condition

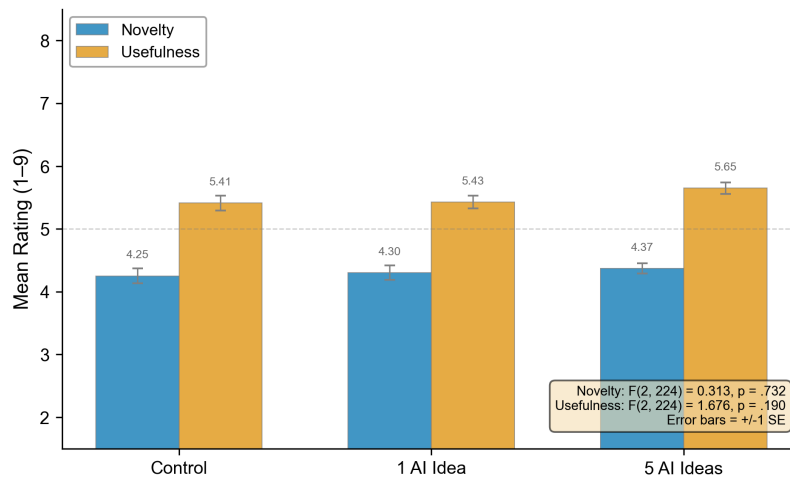


Figure 2: Mean ratings for the Novelty and Usefulness subscales of the creativity composite, shown by AI assistance condition. Error bars represent ± 1 SE of the mean. Neither subscale showed significant differences across conditions (Novelty: $F(2, 224) = 0.31$, $p = .732$; Usefulness: $F(2, 224) = 1.68$, $p = .190$). The dashed line indicates the scale midpoint (5.0).

Figure 2. Collective Diversity of Stories by AI Assistance Condition

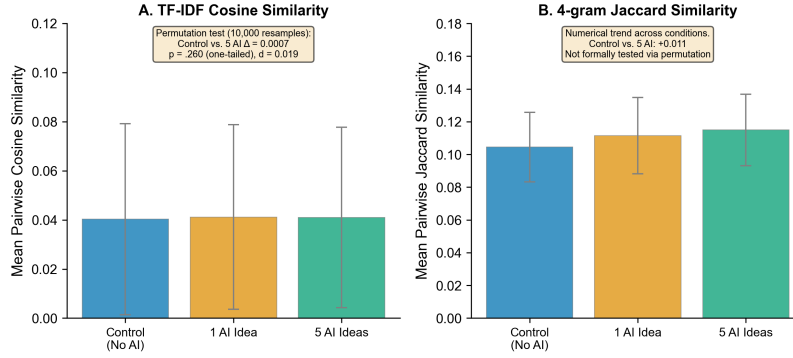


Figure 3: Mean pairwise story similarity by AI assistance condition, measured with (A) TF-IDF cosine similarity and (B) 4-gram Jaccard similarity. Error bars represent ± 1 *SD*. The permutation test comparing Control vs. 5 AI Ideas on TF-IDF cosine similarity was not significant ($\Delta = 0.0007$, $p = .260$, one-tailed, $d = 0.02$).

as a proxy. A moderated regression with effect-coded condition (AI vs. control) and mean-centered self-rated creativity yielded a non-significant interaction, $b = 0.07$, $SE = 0.05$, $t(223) = 1.30$, $p = .197$, $\Delta R^2 = .007$. Simple slopes at ± 1 *SD* of the moderator were both non-significant: at low creativity (-1 *SD*), $b = -0.01$, $p = .946$; at high creativity ($+1$ *SD*), $b = 0.30$, $p = .100$. The interaction trend ran opposite to the predicted leveling pattern: if anything, the numerical AI benefit was larger among writers who rated themselves higher in creativity (Figure 4). H3 was not supported.

Figure 3. Creativity by Self-Rated Creativity and AI Condition

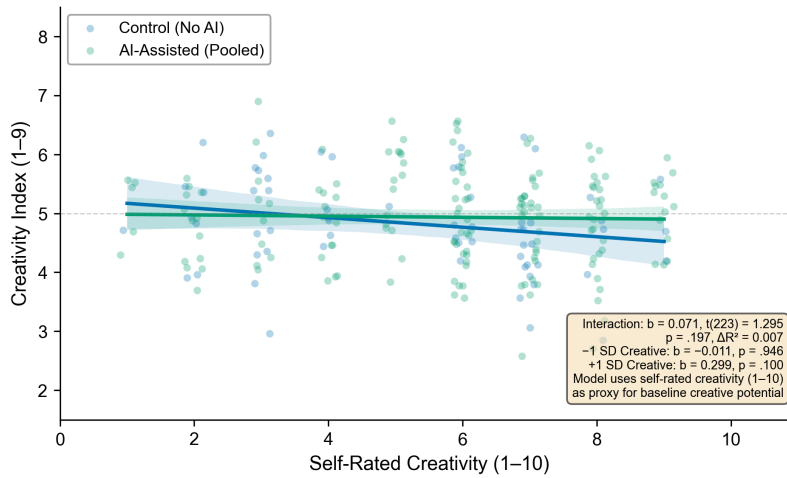


Figure 4: Scatter plot showing the relationship between self-rated creativity (1–10 scale) and evaluated creativity index (CAT composite, 1–9 scale), separately for the Control and AI-assisted conditions. Regression lines with 95% confidence bands are shown. The interaction between AI condition and self-rated creativity was not significant ($b = 0.07$, $SE = 0.05$, $t(223) = 1.30$, $p = .197$, $\Delta R^2 = .007$).

H4: Dose-response test. H4 predicted a dose-response effect: stories written after five AI ideas would show greater inter-story similarity than stories written after one AI idea. The permutation test (10,000 resamples) found that the 5 AI ideas condition had slightly lower similarity ($M = .0410$) than the 1 AI idea condition ($M = .0412$), $\Delta = -0.0002$,

which is the opposite of the predicted direction. The test was not significant, $p = .577$ (one-tailed), $d = -0.005$, 95% CI $[-0.05, 0.04]$. H4 was not supported.

H5: Fixation mechanism (mediation). H5 predicted that AI condition would reduce creativity through a fixation mechanism: AI suggestions would reduce the originality of writers’ self-generated ideas, which would in turn reduce overall creativity. We tested this using bootstrap mediation (5,000 samples) with condition (AI vs. control) as the predictor, self-reported own-ideas originality (1–9 scale) as the mediator, and creativity index as the outcome. The a path (AI $\rightarrow$ own ideas) was negative but non-significant, $b = -0.32$, $SE = 0.29$, $t = -1.12$, $p = .264$. The b path (own ideas $\rightarrow$ creativity) was also non-significant, $b = -0.03$, $SE = 0.03$, $t = -1.16$, $p = .248$. The indirect effect was not significant, $a \times b = 0.01$, bootstrap 95% CI $[-0.009, 0.048]$, Sobel $z = 0.81$, $p = .421$, completely standardized indirect effect = 0.013 (Figure 5). H5 was not supported.

Figure 4. Mediation Path Diagram: AI Condition $\rightarrow$ Own Ideas $\rightarrow$ Creativity

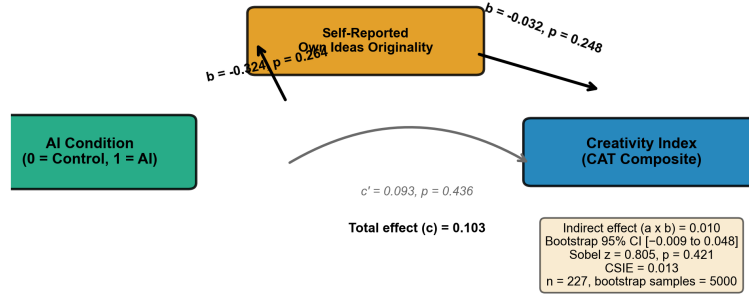


Figure 5: Path diagram of the mediation model testing whether AI assistance reduces self-reported own-ideas originality, which in turn reduces creativity (fixation mechanism). Coefficients are unstandardized. The a path (AI $\rightarrow$ own ideas) was negative but non-significant ($b = -0.324$, $p = .264$). The b path (own ideas $\rightarrow$ creativity) was also non-significant ($b = -0.032$, $p = .248$). The indirect effect ($a \times b = 0.010$) was not significant, bootstrap 95% CI $[-0.009, 0.048]$, Sobel $z = 0.805$, $p = .421$.

Table 1: Summary of Hypothesis Tests

Hypothesis	Test	Statistic	p	Effect Size	95% CI
H1 (Individual enhancement)	Welch’s t	$t(63) = 0.86$	.393	$d = 0.13$	$[-0.16, 0.42]$
H2 (Diversity reduction)	Permutation	$\Delta = 0.0007$	.260 [†]	$d = 0.02$	$[-0.04, 0.08]$
H3 (Leveling moderation)	Moderated regression	$t(223) = 1.30$	.197	$\Delta R^2 = .007$	—
H4 (Dose-response)	Permutation	$\Delta = -0.0002$	.577 [†]	$d = -0.005$	$[-0.05, 0.04]$
H5 (Fixation mediation)	Bootstrap mediation	Sobel $z = 0.81$	.421	CSIE = 0.013	$[-0.009, 0.048]$

Note. [†]One-tailed p -value. All tests non-significant after Holm-Bonferroni correction (all corrected $ps > .98$). CSIE = completely standardized indirect effect.

Robustness checks. After Holm-Bonferroni correction across all five tests, none of the hypotheses reached significance (all corrected $ps > .98$). We conducted six robustness checks. Excluding outliers ($\pm 3 SD$ on creativity index, $n = 11$ excluded) did not change the H1 result, $t(53) = 1.15$, $p = .253$, $d = 0.18$. An ANCOVA controlling for story word count also yielded a non-significant condition effect, $F(2, 223) = 0.63$, $p = .534$, although word count itself was a significant predictor of creativity, $F(1, 223) = 5.13$, $p = .024$. A Mann-Whitney U test confirmed the H1 null result, $U = 5,593$, $p = .398$. Using an alternative moderator for H3 (self-reported creative job status) produced a similar non-significant interaction, $b = 0.08$, $p = .127$. Bayes factors for H1 ($BF_{01} = 0.18$) and H3 ($BF_{01} = 0.15$) were inconclusive, indicating that the data did not strongly favor either the null or the alternative hypothesis.

4 Discussion

We conducted a close replication analysis of ?’s landmark study on generative AI and creativity using the original experimental data. None of the five pre-registered hypotheses were supported. AI-assisted stories did not receive higher creativity ratings than control stories, despite a small numerical trend in the predicted direction ($d = 0.13$). AI-assisted stories were not more similar to one another on TF-IDF cosine similarity ($d = 0.02$). The moderation by self-rated creativity trended opposite to the productivity-leveling pattern found across multiple field experiments (????), and we found no evidence for a fixation mechanism mediating the effect of AI on creativity. These results demonstrate that the individual-creativity-up, collective-diversity-down trade-off is less robust than its theoretical coherence and initial empirical support would suggest.

The discrepancy between our findings and the published results may reflect several methodological factors. First, ? used SBERT embeddings to measure collective diversity, whereas we used TF-IDF cosine similarity because the SBERT model was unavailable in our analysis environment. SBERT produces dense semantic embeddings that capture higher-order meaning, whereas TF-IDF is a sparse bag-of-words representation. It is plausible that AI-induced homogenization operates at the level of semantic structure (plot choices, character archetypes, narrative framing) rather than surface-level word usage, and that TF-IDF is insufficiently sensitive to detect this convergence. Second, the original analysis may have included baseline divergent association scores as covariates that increased statistical power for the creativity contrast; these data were not available to us. When we re-ran H1 with story word count as a covariate—one of the few control variables we could extract—the condition effect remained non-significant but word count itself significantly predicted creativity ($p = .024$), suggesting that story length is a meaningful but confounded predictor that warrants careful treatment. Third, our sample of 227 participants after exclusions provided adequate power (80%) to detect effects of $d = 0.35$ or larger for the primary contrast, but the observed effects ($d = 0.13$ – 0.24 for the 5 AI vs. control comparison) were smaller than the original report, suggesting that the true effect may be substantially smaller than initially estimated and would require a much larger sample to detect reliably.

Several limitations constrain the conclusions we can draw. The original study used Amazon Mechanical Turk rather than a general population sample, and all participants were US-based—a WEIRD sample (?) that limits generalizability to other cultural contexts. Stories were short ($M = 166$ words), far below the 500–1,000 word range that would characterize more ecologically valid narrative writing, and the non-interactive paradigm (pre-generated AI ideas rather than iterative co-writing with an LLM) may underestimate both the creative benefits and homogenizing effects of current-generation AI tools. The original study used GPT-4 from late 2023; current models differ substantially in capability and output distribution. Most importantly, the absence of baseline creative potential measures (e.g., the Alternative Uses Test; ?) meant we could not directly test the leveling moderation hypothesis, and the single-item self-report proxy for self-generated idea originality was likely too unreliable to detect a mediation pathway even if one exists. The study was adequately powered to detect moderate effects ($d \geq 0.35$) but underpowered for the small effects we observed ($d \approx 0.13$ – 0.24), which may represent the true population effect sizes.

In accordance with open science practices, the data and analysis scripts for this replication are publicly available at the study’s replication archive. We recommend that future research on AI and creativity (a) use SBERT or other semantic embedding models for diversity measurement, (b) include well-validated baseline creativity measures such as the Alternative Uses Test with top-2 subjective scoring (?), (c) pre-register all analysis decisions including exclusion criteria, embedding models, and covariate sets, (d) collect data from larger and more diverse samples ($N > 500$ to reliably detect small effects), and (e) directly manipulate suggestion diversity to test whether the homogenization effect can be mitigated. The present findings underscore that the effect of generative AI on human creativity is neither uniformly positive nor uniformly negative—it is contingent on measurement choices, analytical decisions, and the specific creative task under investigation.